

Project Master Journal & Audit Trail (Journal de Bord)

Project: S&P 500 Corporate Governance & Financial Performance Analysis **Course:** Quantitative Data Analysis (M1 – S7, Course code: 2627_ECO_2_EN_009) **Institution:** EM Normandie Business School **Lecturer:** Dr. NGUYEN Anh-Tuan **Lead Researcher:** David & Consulting Project Team **Academic Year:** 2026–2027 **Language:** English (Academic & Executive Standard)

1. Executive Summary & Research Framework

1.1 The Managerial Problematic (Research Question)

Corporate executives, board directors, and ESG fund managers face an ongoing debate: *Does dedicating significant capital and operational focus to institutional corporate governance and social responsibility enhance shareholder value, or does it impose costly bureaucratic frictions that compromise financial profitability?*

To resolve this question empirically, this project establishes a 4-week analytical trajectory centered on:

“Does corporate governance risk (ISS QualityScore) compromise financial profitability and market valuation among S&P 500 firms, or does strong governance provide a resilience and performance premium?”

1.2 The 4-Step Answering Protocol

In accordance with course guidelines (Slide 9 & Slide 41), every hypothesis and statistical result follows a strict four-step protocol:

1. **IDENTIFY:** Justify the statistical test exclusively from the measurement scale of the variables (Nominal, Ordinal, Continuous).
 2. **HYPOTHESISE:** Formulate the Null (H_0) and Alternative (H_1) hypotheses in plain business language.
 3. **DECIDE:** Evaluate the significance level ($\alpha = 0.05$, report exact p -value or $p < .001$) and assess the magnitude/effect size (r , η^2 , Cramér’s V , R^2).
 4. **RECOMMEND:** Provide concrete operational directives answering: *“What does the manager do on Monday morning?”*
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2. Chronological Log of Project Milestones

Date & Time	Phase	Actions Performed	Status
2026-09-21 11:38	Project Inception	Initial virtual environment <code>.venv</code> and preliminary requirements established.	Completed
2026-09-21 11:48	Data Scrape (Initial)	Initial execution of <code>extract_sp500_esg.py</code> pulling 503 S&P 500 tickers via <code>yfinance</code> .	Completed
2026-09-21 18:07	Diagnostic Review	Full audit of course brief <code>AQD_Session1_copie.pdf</code> and existing repository files.	Completed
2026-09-21 18:37	Repository Re-architecture	Git initialized (main branch), directory tree restructured into <code>data/</code> , <code>src/</code> , <code>docs/</code> , <code>figures/</code> , <code>tests/</code> .	Completed
2026-09-21 18:41	Dependency Hardening	Installed verified scientific stack: <code>scipy</code> , <code>statsmodels</code> , <code>matplotlib</code> , <code>seaborn</code> , <code>openpyxl</code> , <code>pytest</code> .	Completed
2026-09-21 18:42	Data Cleaning Pipeline	Automated <code>src/cleaning.py</code> executed: remediated Fiserv (FISV → FI), built Jamovi CSV & Excel.	Completed
2026-09-21 18:42	Automated Testing	Built and executed <code>tests/test_dataset_integrity.py</code> with <code>pytest</code> (6/6 tests passed).	Completed
2026-09-21 18:43	Univariate Engine	Ran <code>src/stats_univariate.py</code> generating 3-centre statistics, dispersion, skewness, kurtosis, Shapiro-Wilk tests.	Completed
2026-09-21 18:43	APA Visualizations	Ran <code>src/visualize.py</code> generating publication-grade figures in <code>figures/</code> .	Completed
2026-09-21 18:45	Deliverable Authoring	Authored <code>Approval_Slip.md</code> , <code>Data_Cleaning_Log.md</code> , <code>Artefact_1_Session_1.md</code> , and Jamovi guide.	Completed

3. Data Provenance, Acquisition & Temporal Grounding

3.1 Data Sources

1. **Constituent Universe:** The list of 503 constituent companies of the S&P 500 index was scraped dynamically from the official Wikipedia directory of S&P 500 companies (maintained from S&P Dow Jones Indices releases).
2. **Corporate Governance Scores:** Sourced from **Institutional Shareholder Services (ISS)** Governance QualityScore ratings accessed via the Yahoo Finance query infrastructure.
 - Scale: Decile scores from 1 (lowest risk, highest governance quality) to 10 (highest risk, poorest governance quality).
 - Core sub-pillars: *Audit Risk*, *Board Structure Risk*, *Compensation Risk*, *Shareholder Rights Risk*, and composite *Overall Governance Risk*.

3. **Financial Fundamentals:** Extracted via Yahoo Finance API (`yfinance`), retrieving audited 10-K SEC regulatory filings.

- Core variables: *Total Revenue* (B), *Market Capitalization* (B), *Net Profit Margin*, *Return on Equity* (ROE), *Market Beta* (5Y monthly), and *Headcount*.

3.2 Why Full Fiscal Year 2025 (FY2025)?

In empirical quantitative finance, using partial current-year figures (e.g. Q1/Q2 2026) creates severe seasonal distortion, annualized extrapolation errors, and interim accounting noise.

- **Audited Completeness:** FY2025 provides complete, 12-month audited financial statements filed with the U.S. Securities and Exchange Commission (SEC).
 - **Macroeconomic Stability:** Captures the full annual economic cycle, ensuring valid comparability across sectors with contrasting seasonality (e.g., Retail peak in Q4 vs. Energy fluctuations).
 - **Synchronicity:** Financial ratios for FY2025 align with Q3/Q4 2025 ISS governance evaluations, ensuring cross-sectional validity.
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4. Comprehensive Data Cleaning & Remediation Audit

4.1 Remediation of Ticker Anomalies: Fiserv (FISV → FI)

- **Diagnosis:** In the raw extract, row 199 (FISV) exhibited missing values across sector, revenue, profit margin, beta, headcount, and governance risk.
- **Root Cause Analysis:** In 2023, Fiserv, Inc. transferred its listing from NASDAQ to the New York Stock Exchange (NYSE) and rebranded its ticker symbol from FISV to FI. The raw extraction script relied on the legacy ticker symbol, causing Yahoo Finance API calls to return null objects.
- **Remediation:** The pipeline programmatically maps FISV to FI, populating its proper sector (*Financial Services* → *Finance*), FY2025 revenue (\$19.09B), profit margin (16.0%), beta (0.88), headcount (42,000), and ISS governance risk scores.

4.2 The Case of Negative Book Equity & Undefined ROE ($N = 32$)

- **Diagnosis:** 32 major blue-chip corporations exhibited missing values (NaN) in the ROE column.
 - ▶ Affected firms include: AbbVie (ABBV), Altria Group (MO), AutoZone (AZO), Booking Holdings (BKNG), Cardinal Health (CAH), Domino's Pizza (DPZ), Fair Isaac (FICO), HCA Healthcare (HCA), Hilton Worldwide (HLT), HP Inc. (HPQ), and Philip Morris.
- **Root Cause (Corporate Finance Theory):**

$$\text{ROE} = \frac{\text{Net Income}}{\text{Shareholders' Equity}}$$

Over the past decade, these mature corporations executed aggressive, debt-financed share repurchase programs (*treasury stock buybacks*). Cumulative buybacks subtracted from retained earnings have driven total book value of equity into negative territory. When book equity is negative, computing a standard ROE yields either a mathematically positive number for loss-making firms or a negative ratio for highly profitable firms—rendering ROE

economically meaningless. Consequently, standard financial databases (including Yahoo Finance and Bloomberg) suppress ROE for negative-equity firms.

- **Remediation & Pedagogical Strategy:**

1. We do **not** arbitrarily impute or drop these corporations; doing so would delete 32 of America's most prominent firms and introduce severe selection bias.
2. We retain ROE for the 471 firms where equity is positive ($N = 471$), while reporting missingness transparently.
3. We introduce **Net Profit Margin** ($\frac{\text{Net Income}}{\text{Revenue}}$) as our co-primary profitability metric. Because revenue is always positive for operating firms, Profit_Margin is 100% defined for all $N = 503$ firms.

4.3 Extreme Outlier Treatment: Masco Corporation (MAS)

- **Diagnosis:** Masco Corp displayed an extreme ROE of 58.625 (5, 862.5%).
- **Root Cause:** Masco held a tiny positive equity balance ($\approx \$80$ M) against nearly \$1B in net profits, causing the ratio denominator to approach zero.
- **Remediation:** In accordance with course guidelines (Slide 21: *"The mean is destroyed by outliers. The median is robust to outliers and skew"*), we preserve the authentic observation without artificial truncation or winsorization. We explicitly report both the mean ($\bar{x} = 39.5\%$) and median ($\text{Mdn} = 16.8\%$) and use the median as the true managerial measure of central tendency.

4.4 Sector Consolidation (Course Constraint: 2 to 5 Categories)

The Global Industry Classification Standard (GICS) specifies 11 raw sectors. To satisfy the mandatory course rule (*Nominal grouping variable MUST have between 2 and 5 categories*), sectors were mapped into 5 homogeneous business groups:

1. **Industrials & Energy** ($n = 148$, 29.4%, **Mode**)
2. **Consumer** ($n = 116$, 23.1%)
3. **Tech & Comms** ($n = 108$, 21.5%)
4. **Finance** ($n = 71$, 14.1%)
5. **Healthcare** ($n = 60$, 11.9%)

5. Summary of Cleaned Dataset Files

1. **data/raw/sp500_esg_dataset_raw.csv:** Pristine original scrape preserved for replication audits.
 2. **data/processed/sp500_esg_cleaned.csv:** Cleaned, imputed, and validated dataset for Python and R pipelines.
 3. **data/processed/sp500_esg_jamovi.csv:** Formatted specifically for Jamovi with standardized short headers (Ticker, Company, Sector, Revenue_B, etc.).
 4. **data/processed/sp500_esg_jamovi.xlsx:** Excel workbook featuring sheet 1 (Data) and sheet 2 (Codebook_Jamovi) with explicit variable definitions, measure types, and measurement levels.
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6. Git Version Control Log

The project repository is under strict local Git version control:

- **Repository:** Group Work S&P 500
- **Active Branch:** main
- **Initial Commit:** Project baseline structure and raw data assets.
- **Pipeline Commit:** Modular data cleaning, univariate engine, APA visualizations, automated test suite, and submission documentation.